EE1007
Introduction to Computer Science Laboratory

Lab 4: Control Statement II (Repeat Control)

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P110009: Lab 4.1 Prime Factorization

- In this program, you need to write a program that can decompose the prime factors of any positive integer
- Then, the prime factors are displayed in increasing order
- The program need to design fool-proof to avoid program errors
- Please use **if… else**, **while** or **do… while** to implement your program, otherwise you will obtain 0 point
- The range of each integer is $[2 \sim 2^{31}-1]$

P110009: Lab 4.1
Prime Factorization

◆The screen dialog should appear as follows:

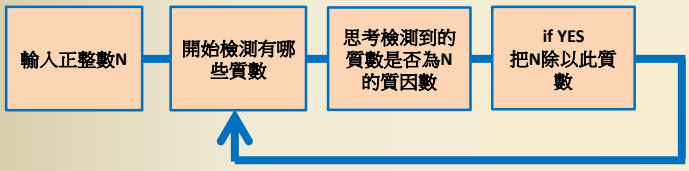
```
Enter a number: -222
Please enter a correct number:1
Please enter a correct number:1260
1260 = 2*2*3*3*5*7

Process returned 0 (0x0)   execution time : 11.066 s
Press any key to continue.
```


LAB4.1

◆計算質因數

◆思考方向



LAB4.1

◆檢測質數

```
while(i<=number){
    int x=1,check=0;
    //檢測質數
    while(i<=x){
        if(i%x==0){
            check++;
        }
        if(check>2){
            break;
        }
    }
    if(check>2){
        i++;
        continue;
    }

    i++;
    if(number<=1){
        break;
    }
}
```

LAB4.1

◆判斷是否為N的因數

```
//質因數分解
while(number%i==0){
        number/=i;
    if(number<=1){
        cout<<i<<endl;
    }
    else{
        cout<<i<<"*";
    }
}
```

P110010: Lab 4.2
Weekly calculation

- In this problem, you need to write a program which can find the corresponding day of a given date
- 2021.01.01 is Friday
- The program need to design fool-proof to avoid program errors
- Please use **switch…case** to complete this program, **do not use if…else** otherwise no score will be given.
- The range of each integer is [20210101~2^{63}-1].

Lab 4.2
Weekly calculation

- In this program, you **cannot** use any formulas to calculate the day.
- For example,
  - 基姆拉爾森計算公式
    W = (d + 2 * m + 3 * (m + 1) / 5 + y + y / 4 - y / 100 + y / 400) mod 7
  - 蔡勒公式
    W = [C/4] - 2C + y + [y/4] + [13 * (M+1) / 5] + d - 1

Lab 4.2
Weekly calculation

◆The screen dialog should appear as follows:

```
Enter a date (ex. 20211014): 20200101
This date does not exist or is within the calculation range. Please enter a correct date (ex. 20211014): 20223113
This date does not exist or is within the calculation range. Please enter a correct date (ex. 20211014): 20211014
20211014 is Thursday.

Process returned 0 (0x0)   execution time : 32.933 s
Press any key to continue.
```


LAB4.2

◆判斷星期

◆思考方向



LAB4.2

◆切割年月日

```
long long int ay,am,ad,by,bm,bd;
ay=date/10000;
am=date/100-ay*100;
ad=date-am*100-ay*10000;
by=base/10000;
bm=base/100-by*100;
bd=base-bm*100-by*10000;
```

LAB4.2

◆確認輸入的年月日是否正確

➢判斷正確的月份天數

```
int maxD;
switch(ay%4){
    case 0:
        switch(am){
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            maxD=31;
            break;
        case 2:
            maxD=29;
            break;
        case 4: case 6: case 9: case 11:
            maxD=30;
        }
        break;
    default:

        switch(am){
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            maxD=31;
            break;
        case 2:
            maxD=28;
            break;
        case 4: case 6: case 9: case 11:
            maxD=30;
        }
        break;
}
```

年月的條件

```
while(ay<2021 || am<1 || am>12 || ad<1 ||ad>maxD){
}
```

LAB4.2

◆計算輸入的年月日跟20210101的差幾天

```
for(int i=by;i<ay;i++){
    switch(i%4){
        case 0:
            dif+=366;
            break;
        default:
            dif+=365;
            break;
    }
}
```

```
for(int i=bm;i<am;i++){
        // cout<<i<<endl;
    switch(ay%4){
    case 0:
        switch(i){
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            dif+=31;
            break;
        case 2:
            dif+=29;
            break;
        case 4: case 6: case 9: case 11:
            dif+=30;
        }
        break;
    default:

        switch(i){
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            dif+=31;
            break;
        case 2:
            dif+=28;
            break;
        case 4: case 6: case 9: case 11:
            dif+=30;
        }
        break;

    }
}
```

Evaluation

◆Lab4.1(40%)

- 5 testcases, 8% each

◆Lab4.2(60%)

- 5 testcases, 12% each





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